

A-LIS DB Backup SOP - Linux/Ubuntu

Version: 2.0

Audience: ICT Officers / A-LIS Implementers

Estimated Setup Time: 15–20 Minutes | **Created By:** Benson Okello | **Authored:** 10th July 2026

Objective

This guide describes how to configure an A-LIS installation to automatically upload compressed database backups to the UNHLS Central Backup Server, and how to restore from those backups if a facility's database instance is lost or corrupted.

Once the backup setup is completed:

- Database backups will run automatically.
- Backups will be copied off-site every day.
- Local backups older than three days will be removed automatically.
- No further user intervention is required.

Prerequisites

Before starting, ensure you have:

- SSH access to the A-LIS server.
- MySQL database credentials.
- Internet connectivity to the Central Backup Server.

“Once you have all these in place, scroll to the bottom for a quick Motivational Quote”

Step-by-Step Backup Process

1. Create Backup Directory & SSH Key on ALIS server

Login to the Facility's A-LIS server and Create a backup directory.

i. Create Backup Directory

```
mkdir ~/.backup
```

ii. While still on the A-LIS facility server generate a dedicated Ed25519 SSH key for the A-LIS off-site backup process:

```
ssh-keygen \  
-t ed25519 \  
-f ~/.ssh/alis_backup_ed25519 \  
-C "alis-offsite-backup"
```

iii. When prompted for Passphrase, Press **ENTER**:

Enter passphrase (empty for no passphrase):

Leave the passphrase empty and press **ENTER**.

When prompted again:

Enter same passphrase again:

iv. Protect the private SSH key:

```
chmod 600 ~/.ssh/alis_backup_ed25519
```

vi. Verify that the keys were created:

```
ls -lah ~/.ssh/alis_backup_ed25519*
```

The following files should exist:

```
~/.ssh/alis_backup_ed25519  
~/.ssh/alis_backup_ed25519.pub
```

v. Display the public key:

```
cat ~/.ssh/alis_backup_ed25519.pub
```

Example output:

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIxxxxxxxxxxxxxxxxxxxxxxxx alis-offsite-backup
```

2. Register the Public Key & Download Backup Script From The CIS Portal

i. Log into the CIS Portal <https://support.cphl.go.ug/>

- Click Staff Login

- Enter your email and password to log in
- On the right hand pane, under **Manager Lists** , select **Infrastructure: A-LIS Backup & Configuration**

CPHL ICT SUPPORT PORTAL

Anti-Spam Infrastructure ▾ Track Ticket **Home** Logout

HOME

Welcome, CIS Administrator Logout

Overview **Support Team Performance** Manager Dashboard

6
Assigned Tickets

0
In Progress

2
Resolved

2
SLA Compliant

Recent Tickets View All

TICKET	SYSTEM	STATUS	DATE LOGGED	ACTIONS
CIS-260812-001	ALIS	Assigned	2026-08-12	View
CIS-260804-001	SIMCS	Resolved	2026-08-04	View
CIS-260731-001	OUTBREAK LIMS (O-LIMS)	Assigned	2026-07-31	View
CIS-260730-001	IRRDS	Assigned	2026-07-30	View
CIS-260725-001	IRRDS	Closed	2026-07-25	View
CIS-260722-001	VL Dashboard	Resolved	2026-07-22	View

Administration

Manage ICT Users

Manager Lists

Infrastructure: A-LIS Backups & Configurations

A-LIS Key Update Reasons

Systems

Modules

Regions

Facilities

Departments

Designations

Issue Types

Priority/Impact Levels

Ticket Statuses

Resolution Categories

Closure Reasons

- Click **Add Backup Configuration** and fill in the form that Opens. Copy & paste the generated Key in **Step1 v** above and Save

CPH ICT SUPPORT PORTAL

Anti-Spam Infrastructure Track Ticket Home Logout

INFRASTRUCTURE

A-LIS Backups & Configuration

Backup Configuration

0 Pending

Search Backup Configuration

Search by facility, district, region

FACILITY
Anaka Hospital Nwaya Mid Northern ANAKAHOSPI
Gulu R R Hospital Gulu Mid Northern GULURRHOSP
Abim Hospital Abim North East ABIMHOSPIT
Jinja R R Hospital Jinja East Central JINJARRHOS
Mukono General Hospital Mukono Central 2

Add Backup Configuration

Save the facility backup settings, provision the central directory, and generate the facility backup script.

Backup Restoration SOP
Download the Linux Ubuntu off-site database backup and restoration SOP for step-by-step guidance.

Download SOP

Region *
Select region

District *
Select district

Used only to narrow the facility list. Choose a region first.

Facility *
Select facility

Choose a district first.

Backup Directory Name *
Generated automatically from the selected facility and validated again on save.

A-LIS Database Name *
alis_db

A-LIS Database Username *
alis_user

A-LIS Database Password *
Enter database password

SSH Public Key *
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAA... alis-offsite-backup

Cancel Save Configuration

- Click the **Download** button on the corresponding Facility Key to download the **backupscrip.sh** file which you will copy to the `~/backup` directory you created in step one.

FACILITY	ADDED BY	DATE ADDED	BACKUP DIRECTORY	STATUS	ACTION
Bugambo H/C II Luuka East Central BUGAMBOHCII	CIS Administrator	2026-07-09 11:26	bugambo_hcii	PROVISIONED	Download Update Deactivate
Kaweweta H/C III Nakaseke District North Buganda KAWEWETA03	CIS Administrator	2026-07-09 14:33	kaweweta_hciiii	DISABLED	Update Activate

ii. Move the downloaded script into the backup directory created in Step 1:

```
mv ~/Downloads/backupscrip.sh ~/backup/
```

iii. Make the script executable:

```
sudo chmod +x ~/backup/backupscrip.sh
```

iv. Change permissions on the `~/backup` directory:

```
sudo chmod -R 777 ~/backup/
```

3. Test the Backup Set up

i. In the terminal, run:

```
~/backup/backupscrip.sh
```

You should see:

```
Connected to 192.168.1.1 (example IP address, you will see the Remote Server IP here)
Uploading...
```

Check the CISPortal for the Last Backup Column status for the facility you have worked on. **No backup found** means it did not succeed, a datetime stamp confirms it was successful.

China Uganda Friendship Hospital-Naguru Kampala Kampala CHINAUGAND	Kabogoza Blasio	2026-07-27 08:51	china_uganda_friendship_hospital_naguru	No backup found	● PROVISIONED	Download Update Deactivate
Kawolo Hospital Buikwe Central 2 KAWOLOHOSP	Benson Okello	2026-07-25 10:49	kawolo_hospital	2026-08-11 21:02 22 hours ago 33.2 MB	● PROVISIONED	Download Update Deactivate

4. Automate Back up process

Note: If you want this backup to happen automatically at regular intervals, you can add the script to `cron` to schedule backups.

i. Open cron

```
crontab -e
```

ii. Add a cron job to run the script daily at midnigt (adjust as needed):

```
#makes database dump everyday at midnight
0 0 * * * ~/.backup/backupscrip.sh >> ~/backup_cron.log 2>&1
```

Motivational Quote:

**“ If it does not work on your computer, tell them it worked on mine.
Or, rather just cry**

Restoring from a Backup

Use this section when a facility's A-LIS database instance is broken, corrupted, or lost, and needs to be restored from the off-site gzip backups produced by the process above.

1. Retrieve the Backup File

Connect to the central backup server and pull down the facility's most recent backup (or an earlier one, if the latest is suspected to be affected):

```
sftp -i ~/.ssh/alis_backup_ed25519 -P 22 backupuser@105.27.247.146

cd /dumps/<facility>

ls -la

get alis_db_name_2026-07-08_02-00-00.sql.gz

bye
```

Note: Replace <facility> and the filename with the actual facility folder and backup you need.

2. Verify the Backup File Integrity

Before touching the live database, confirm the archive isn't corrupted:

```
gzip -t alis_db_name_2026-07-08_02-00-00.sql.gz && echo OK || echo CORRUPT
```

If the check reports CORRUPT, retrieve an earlier dated backup from the same facility folder and repeat this check.

3. Safety-Dump the Existing Database (if it's still reachable)

If the current instance is still partially running, take one more dump before overwriting anything, in case the "broken" instance still holds data worth keeping:

```
mysqldump -u alis_db_username -p alis_db_name > pre_restore_safety_dump.sql \

2>/dev/null || echo "skip — instance unreachable"
```

4. Recreate the Database (if needed)

If the database no longer exists on the target MySQL instance:

```
mysql -u root -p -e "CREATE DATABASE IF NOT EXISTS alis_db_name;"
```

5. Restore the Backup

```
gunzip -c alis_db_name_2026-07-08_02-00-00.sql.gz | mysql -u alis_db_username -p alis_db_name
```

This decompresses the backup and pipes the SQL directly into MySQL. Depending on database size, this can take a few minutes.

6. Confirm the Restore

- Log in and spot-check row counts on a couple of key tables.
- Open the A-LIS application itself and confirm it connects and loads recent records.
- Check the application logs for connection or schema errors.

Restoration Notes

- If MySQL will not start at all (not just a corrupted database), fix or reinstall MySQL first, confirm `mysql -u root -p` works, then proceed with the restore.
- If restoring to a different server than the original, check MySQL version compatibility — routines and triggers from a newer MySQL version can fail to import on an older one.
- Test this restore procedure once on a non-production machine so the steps are familiar before they're ever needed under pressure.

🕒 Revision #33

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